

Source Reference List

Every source cited while teaching Module 8 (Data Preparation for ML), organized by topic and subtopic, with a note on what each source is useful for.

MODULE 8: Data Preparation for ML

Topic 1: Data Preparation Fundamentals

1a. Why data preparation matters

<https://www.oreilly.com/library/view/hands-on-machine-learning/9781098125967/>

For: Hands-On ML by Géron, Ch. 2 — data cleaning section

<https://developers.google.com/machine-learning/crash-course/representation/cleaning-data>

For: Google ML Crash Course — why cleaning data matters

<https://scikit-learn.org/stable/modules/preprocessing.html>

For: Scikit-learn's overview of preprocessing

1b. Raw vs processed data

<https://developers.google.com/machine-learning/crash-course/representation/feature-engineering>

For: Google ML Crash Course — feature engineering overview

https://pandas.pydata.org/docs/user_guide/missing_data.html

For: pandas — handling raw/messy data with missing values

1c. Data quality challenges

<https://developers.google.com/machine-learning/data-prep/construct/collect/data-quality>

For: Google ML Crash Course — data quality challenges

<https://www.ibm.com/topics/data-quality>

For: IBM — what data quality is and why it matters

1d. ML-ready datasets

<https://scikit-learn.org/stable/modules/preprocessing.html>

For: Scikit-learn — preprocessing to reach ML-ready form

<https://developers.google.com/machine-learning/crash-course/representation/feature-engineering>

For: Google ML Crash Course — feature engineering for readiness

https://pandas.pydata.org/docs/reference/api/pandas.get_dummies.html

For: pandas — `get_dummies()` for encoding to ML-ready form

1e. Data preparation lifecycle

<https://www.ibm.com/docs/en/spss-modeler/saas?topic=dm-crisp-help-overview>

For: IBM — CRISP-DM lifecycle overview

<https://developers.google.com/machine-learning/data-prep>

For: Google ML Crash Course — data preparation lifecycle

<https://www.oreilly.com/library/view/hands-on-machine-learning/9781098125967/>

For: Hands-On ML by Géron, Ch. 2 — end-to-end ML project workflow

Topic 2: Data Cleaning

2a. Missing values

https://pandas.pydata.org/docs/user_guide/missing_data.html

For: pandas — working with missing data

<https://scikit-learn.org/stable/modules/impute.html>

For: Scikit-learn — imputation of missing values

<https://developers.google.com/machine-learning/data-prep/construct/collect/data-quality>

For: Google ML Crash Course — data quality (missing data)

2b. Duplicate records

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.duplicated.html>

For: pandas — DataFrame.duplicated() reference

<https://developers.google.com/machine-learning/data-prep/construct/collect/data-quality>

For: Google ML Crash Course — data quality (duplicates)

2c. Invalid entries

<https://developers.google.com/machine-learning/data-prep/construct/collect/data-quality>

For: Google ML Crash Course — data quality (invalid entries)

https://pandas.pydata.org/docs/user_guide/indexing.html

For: pandas — boolean indexing for detecting invalid values

2d. Outlier identification

<https://developers.google.com/machine-learning/data-prep/transform/normalization>

For: Google ML Crash Course — outliers and normalization

https://scikit-learn.org/stable/modules/outlier_detection.html

For: Scikit-learn — outlier detection methods

<https://www.statisticshowto.com/probability-and-statistics/interquartile-range/>

For: Statistics How To — IQR method explained

2e. Data consistency improvements

https://pandas.pydata.org/docs/user_guide/text.html

For: pandas — string methods for cleaning/standardizing text

<https://developers.google.com/machine-learning/data-prep/construct/collect/data-quality>

For: Google ML Crash Course — data quality (consistency)

Topic 3: Data Transformation

3a. Feature transformation

<https://developers.google.com/machine-learning/crash-course/representation/feature-engineering>

For: Google ML Crash Course — feature transformation overview

<https://scikit-learn.org/stable/modules/preprocessing.html>

For: Scikit-learn — preprocessing techniques overview

3b. Scaling

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

For: Scikit-learn — MinMaxScaler reference

<https://developers.google.com/machine-learning/data-prep/transform/normalization>

For: Google ML Crash Course — scaling techniques

3c. Normalization

<https://scikit-learn.org/stable/modules/preprocessing.html#normalization>

For: Scikit-learn — normalization reference

<https://developers.google.com/machine-learning/data-prep/transform/normalization>

For: Google ML Crash Course — normalization techniques

3d. Encoding categorical variables

<https://scikit-learn.org/stable/modules/preprocessing.html#encoding-categorical-features>

For: Scikit-learn — encoding categorical features

https://pandas.pydata.org/docs/reference/api/pandas.get_dummies.html

For: pandas — get_dummies() for one-hot encoding

3e. Data standardization

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

For: Scikit-learn — StandardScaler reference

<https://developers.google.com/machine-learning/data-prep/transform/normalization>

For: Google ML Crash Course — standardization vs normalization

Topic 4: Feature Preparation

4a. Feature creation

<https://developers.google.com/machine-learning/crash-course/representation/feature-engineering>

For: Google ML Crash Course — creating new features

https://pandas.pydata.org/docs/user_guide/timeseries.html

For: pandas — working with dates/times for derived features

4b. Feature selection awareness

https://scikit-learn.org/stable/modules/feature_selection.html

For: Scikit-learn — feature selection methods

<https://developers.google.com/machine-learning/crash-course/representation/feature-engineering>

For: Google ML Crash Course — selecting useful features

4c. Removing irrelevant information

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop.html>

For: pandas — DataFrame.drop() reference

<https://developers.google.com/machine-learning/crash-course/representation/feature-engineering>

For: Google ML Crash Course — removing unhelpful columns

4d. Improving model readiness

<https://scikit-learn.org/stable/modules/preprocessing.html>

For: Scikit-learn — preprocessing overview (readiness checks)

<https://developers.google.com/machine-learning/data-prep>

For: Google ML Crash Course — data preparation checklist

4e. Data quality improvements

<https://developers.google.com/machine-learning/data-prep/construct/collect/data-quality>

For: Google ML Crash Course — data quality, tied to features

Topic 5: Data Integration

5a. Combining datasets

https://pandas.pydata.org/docs/user_guide/merging.html

For: pandas — concatenation and merging basics

<https://developers.google.com/machine-learning/data-prep>

For: Google ML Crash Course — combining data sources

5b. Data merging

https://pandas.pydata.org/docs/user_guide/merging.html

For: pandas — merge, join, concatenate reference

5c. Data aggregation

https://pandas.pydata.org/docs/user_guide/groupby.html

For: pandas — GroupBy: split-apply-combine reference

5d. Working with multiple sources

https://pandas.pydata.org/docs/user_guide/merging.html

For: pandas — aligning data before merging

<https://cloud.google.com/learn/what-is-data-integration>

For: Google Cloud — challenges of multi-source data integration

5e. Dataset consistency

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.merge.html>

For: pandas — DataFrame.merge() validate parameter

Topic 6: Data Pipeline Thinking

6a. Data ingestion

https://pandas.pydata.org/docs/user_guide/io.html

For: pandas — IO tools for reading data sources

<https://cloud.google.com/learn/what-is-data-ingestion>

For: Google Cloud — what data ingestion is

6b. Data cleaning workflows

https://pandas.pydata.org/docs/user_guide/basics.html#function-application

For: pandas — function application for reusable workflows

6c. Transformation workflows

<https://scikit-learn.org/stable/modules/compose.html>

For: Scikit-learn — Pipeline for chaining transformations

6d. Reusable processing pipelines

<https://scikit-learn.org/stable/modules/compose.html>

For: Scikit-learn — Pipeline and FeatureUnion reference

6e. Automation awareness

<https://airflow.apache.org/docs/apache-airflow/stable/core-concepts/overview.html>

For: Apache Airflow — core concepts for pipeline scheduling

<https://cloud.google.com/learn/what-is-a-data-pipeline>

For: Google Cloud — what a data pipeline is

Topic 7: Data Leakage Awareness

7a. What is data leakage

<https://developers.google.com/machine-learning/testing-debugging/common/data-errors>

For: Google ML Crash Course — data leakage explained

https://scikit-learn.org/stable/common_pitfalls.html#data-leakage

For: Scikit-learn — common pitfalls: data leakage

7b. Common leakage scenarios

https://scikit-learn.org/stable/common_pitfalls.html#data-leakage

For: Scikit-learn — types of data leakage scenarios

<https://developers.google.com/machine-learning/testing-debugging/common/data-errors>

For: Google ML Crash Course — common data errors

7c. Avoiding invalid datasets

https://scikit-learn.org/stable/common_pitfalls.html#data-leakage

For: Scikit-learn — correct order of operations to avoid leakage

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html

For: Scikit-learn — train_test_split reference

7d. Reliable ML workflows

<https://scikit-learn.org/stable/modules/compose.html>

For: Scikit-learn — Pipeline for leakage-safe workflows

7e. Validation awareness

https://scikit-learn.org/stable/modules/cross_validation.html

For: Scikit-learn — cross-validation reference

<https://developers.google.com/machine-learning/crash-course/validation>

For: Google ML Crash Course — validation concepts

Note: some sources repeat across subtopics because a single reference page (e.g. a full library documentation page) covers several related concepts at once.